



The Hidden Palette:

Uncovering Fashion Shopping Trends

2025

RLV

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Problem / Goal

How do **fashion attributes**
co-occur, and what do they
reveal about **shopping trends**?

- Treat each product like a bundle of choices (**season, gender, color, item type**).
- Look for hidden palettes that shape how people shop.
- Identify patterns that aren't obvious at the item-level but emerge across the dataset.



Dataset [Fashion Product Images]!

Source: Kaggle

Size: 44,424



- Key Features: **gender, article type, base color, season**
- Intentionally excluded broader/redundant columns (**masterCategory, subCategory, year**)
- Each row in the dataframe is treated as a transaction, each column an item
- Example: { 'fall', 'men', 'navy blue', 'shirts' }

styles.csv (4.33 MB)

Detail Compact Column

▲ gender	▲ articleType	▲ baseColour	▲ season
Men	Shirts	Navy Blue	Fall
Men	Jeans	Blue	Summer
Women	Watches	Silver	Winter
Men	Track Pants	Black	Fall
Men	Tshirts	Grey	Summer
Men	Tshirts	Grey	Summer
Men	Shirts	Green	Summer
Women	Shirts	Purple	Summer
Men	Socks	Navy Blue	Summer
Men	Watches	Black	Winter

Techniques



Apriori

Finding attribute combinations that frequently appear together.



Season Splits

Diving deeper into seasonal trends. What happens in the summer, winter, spring, and fall?



Dimensionality Reduction

Using MCA/PCA to better reveal how attribute levels relate to one another



Clustering

Group products into shopper-like style segments. Are there certain types of customers?



Key Findings

Summer consists of more casual wear, while Winter consists of more accessories. Fall is dominated by men and basics for layering. On the other hand, Spring shows more scents and brighter colors.



Key Findings

Clear seasonal marketing trends emerge between seasons. Men & women shop differently. Light apparel dominates summer, watches/accessories and dark colors trend in the winter. Fragrance in the spring.



Key Findings

Products naturally group by season, gender, and color combinations. Season is the most influential organizing factor.



Key Findings

Season and gender are dominant organizing dimensions. Products naturally group who they target and when they're worn.

Seasonal Trends

Winter

men,
watches

black, men,
watches

Watches,
accessories, and
dark colors: black,
brown, blue

heels, women

black, men

black,
watches

handbags, women

Women: 54%
Men: 38%

Summer

men,
tshirts

men, blue

T-shirts dominate.
Colors: black, blue,
white

black, men

men, casual
shoes

Women: 39%
Men: 51%

men,
white

men, shirts

Spring

deodorant,
men

women, lipstick

Fragrance,
deodorant, nail
polish, lipstick
Color: brown, pink,
black

men, perfume and body mist

nail polish, women

women,
pink

Women: 64%
Men: 32%

Fall

men,
white

black

Fall shirts &
layering pieces in
blues, blacks, and
white

men, shirts

men, tshirts

men,
blue

women, kurtas

Women: 32%
Men: 62%

Customer Segmentation

Men's Winter Accessories

Women's Summer/Spring Apparel

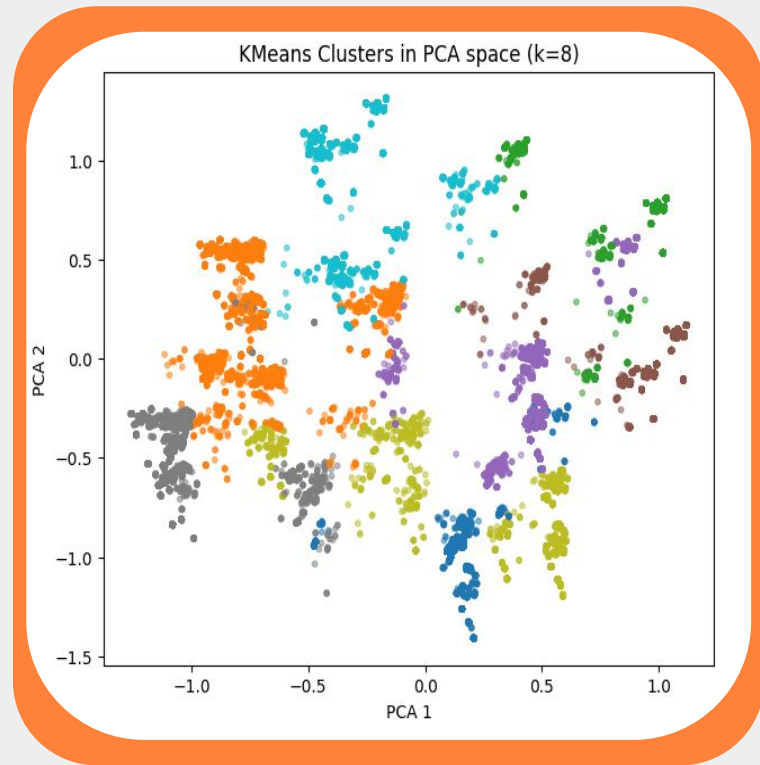
Men's Summer Basics

Women's Winter Fashion & Accessories

Men's Summer Casual & Sportswear

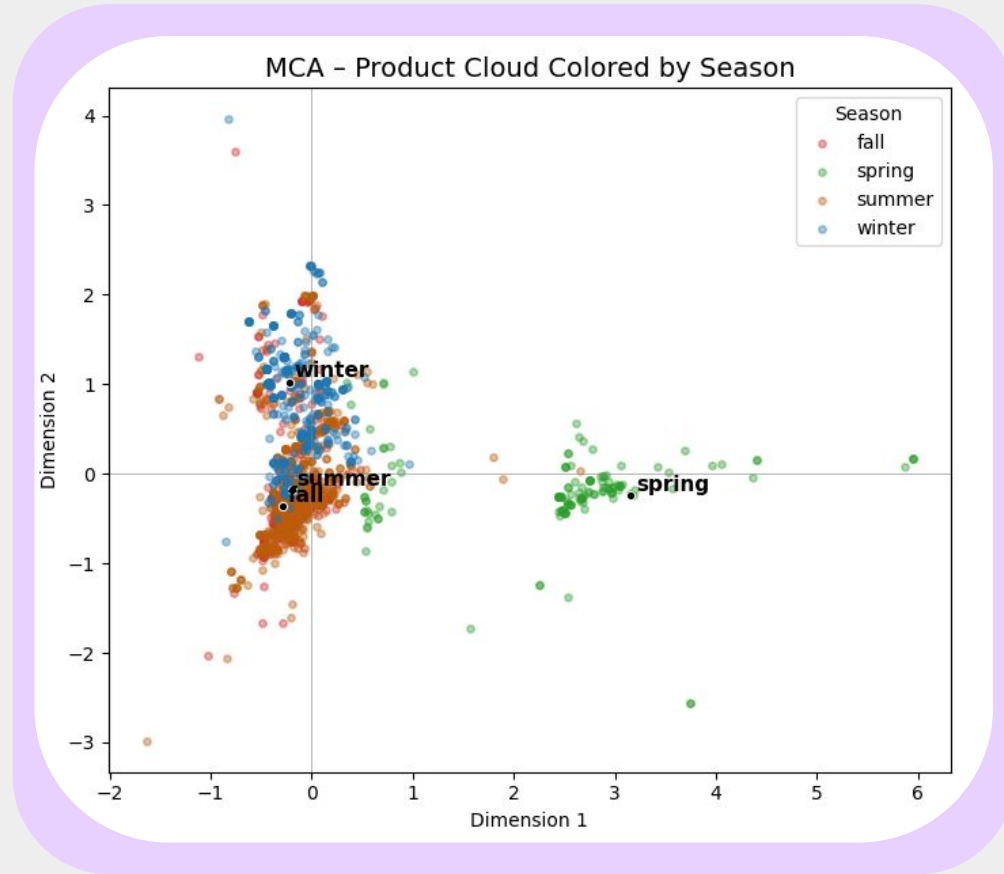


Season & gender are dominant organizing dimensions. Products naturally group by **who** they target and **when** they're worn.



KEY INSIGHT:

Consumer **Focus & Product**
Design typically shift
dramatically with the **seasons**.



Why These Trends Matter

- Trend patterns form predictable palettes along season + gender + color
- Useful for:
 - Seasonal marketing
 - Inventory planning
 - Smarter product recommendations
- Clear gendered style clusters show how products naturally align with shopper preferences
- Identifying distinct style groups (summer basics, winter accessories, etc) helps structure recommendations and inventory planning
- Understanding how items co-occur uncovers the hidden logic of fashion shopping

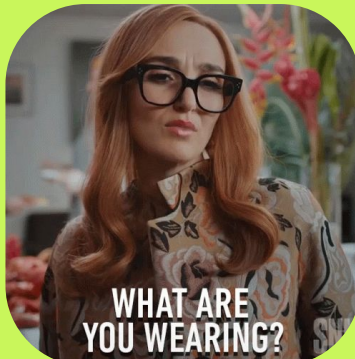


Final Threads



Interpretable Fashion Clusters

Dimensionality Reduction + K-Means reveal meaningful product groupings



Season + Gender Drive What We Wear

It's all about who, what, and wear



Seasonality Shapes Product Types

Seasonality and product type form the strongest structural patterns

Future

Choi, Collis,
Venkataraghavan

We uncovered the hidden rhythms of fashion.
Seasonal shifts, color habits, & style patterns.

- Dataset with broader time period to see how trends shift throughout years/decades
- Use the images in the dataset to analyze textures, fabrics, and silhouettes.
- Source Dataset with attributes like [age, brand, price] to further classify and cluster segmentation of *who, what, "wear"*

